# Alex Zhang

alexzhang2017@gmail.com | +61 402 771 360 | alz11.com | linkedin.com/in/alexzhang7 | github.com/AlZ11

## Education

**Monash University** — Melbourne, Australia
*Bachelor of Computer Science (Distinction); Data Science Specialization* — *Feb. 2023 – Oct. 2026*

**Waseda University** — Tokyo, Japan
*Bachelor of Engineering, Computer Science & Communications Engineering (Exchange)* — *Apr. 2026 – Sep. 2026*

**University of Texas at Austin** — Austin, USA
*Bachelor of Science, Mathematics (Exchange)* — *Aug. 2024 – Dec. 2024*

## Technical Skills

**Programming Languages**: Python, C++, Java, JavaScript, SQL
**Frameworks & Libraries**: NumPy, Pandas, React, Tailwind CSS, Spring Boot
**Infrastructure & Tools**: AWS, Selenium, Git, Linux, CI/CD
**Languages**: English (Native), Chinese Mandarin (Native), Japanese (JLPT N4)

## Experience

**AI Research Intern** — Sep. 2026 – Present
*CSIRO* — *Melbourne, Australia*

- Currently architecting an LLM-based privacy audit pipeline that checks whether mHealth apps' privacy claims match their actual behaviour, by cross-checking privacy policies, Google Play Data Safety labels, APK source code and network traffic. Extending a prexisting CSIRO assessment framework.
- Designing a multi-agent LLM orchestration layer to evaluate applications against the 50 distinct cybersecurity criteria of the ACSIMA framework, Australian relevant privacy standards. Isolating metadata validation from legal text analysis to enforce strict context boundaries and prevent model hallucinations.
- Co-authoring "Mobile App Privacy Policy Analyzer" paper supervised by Dr Chamikara Arachchige

**AI Intern** — Aug. 2025 – Dec. 2025
*Hawthorn FC* — *Melbourne, Australia*

- Engineered AI agents to process GPS and match-event telemetry, reducing data processing latency by **90%** and eliminating manual workflows.
- Streamlined the opposition analysis process by developing scripts to scrape and aggregate public league data, providing coaches with instant access to competitor metrics.

## Projects

**NASDAQ Feed Handler**

- Engineered a low-latency market data feed handler processing **11.5 million msg/s** on a single core.
- Implemented a Zero-Copy architecture using Memory Mapped Files (mmap) and reinterpret_cast to eliminate user-space buffer copies.
- Optimized Order Book data structures by replacing Red-Black Trees (std::map) with contiguous memory vectors (std::vector) for cache locality, improving throughput by **57%**.

## Leadership & Awards

**Accenture** | *Tech Bootcamp First Place Winner – Out of 10 Teams (100+ Students)* — Feb. 2026
**WIRED Monash** | *Technical Events Lead – Ran 6 Workshops for 150+ Students* — Oct. 2024 – May. 2025
**Australia-China Youth Association** | *Case Competition Finalist–Out of 50 Teams (200+ Students)* — Mar. 2024